

Shizuka Takao

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EDUCATION

University of California, Irvine

B.S. in Computer Science, Intelligent Systems Specialization, GPA: 3.98

Irvine, CA

Expected June 2027

Diablo Valley College

A.S. in Computer Science and Mathematics, GPA: 4.00

Pleasant Hill, CA

Graduated May 2024

Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Artificial Intelligence, Machine Learning and Data-Mining, Application of Probability in CS, Information Retrieval, Introduction to Data Management, Principles of System Design, Probabilistic Graphical Models

EXPERIENCE

Undergraduate Research Assistant

Jun 2025 – Present

Ding's Lab, UC Irvine

Irvine, CA

- Developed a cytoplasm segmentation pipeline combining nuclei localization using GroundingDINO, nuclei-seeded watershed, and SAM box prompting to generate single-cell masks from limited labeled data
- Prepared Cellpose-SAM training data from MATLAB tracking annotations, handling unlabeled regions, and organizing train/validation/test splits to reduce data leakage
- Benchmarked Cellpose-SAM against the watershed+SAM baseline using IoU, Dice, and frame-level evaluation metrics to assess segmentation accuracy and robustness across multi-channel FISH images
- Built a Tkinter GUI with annotation, brush/eraser editing, export, and multithreaded segmentation, cutting manual segmentation time by 90%

Data Science Fellow

Aug 2025 – Dec 2025

American Express

Remote

- Built an end-to-end credit risk classification pipeline spanning data cleaning, feature engineering, class imbalance handling, and hyperparameter tuning
- Trained and evaluated XGBoost models for imbalanced classification, achieving 0.83 validation accuracy
- Developed a Streamlit application for real-time credit score probability inference

Software Engineer Intern

Jun 2023 – Aug 2023

Kasanare Corporation

Tokyo, Japan

- Developed LLM-powered chatbots using retrieval-augmented generation pipelines built with LangChain and Pinecone
- Applied zero-shot and chain-of-thought prompting strategies to improve response grounding and retrieval relevance

PROJECTS

Sokoban Solver | Python, PyTorch, Gymnasium

Apr 2026 – Jun 2026

- Developed a Sokoban-solving framework comparing BFS, Greedy Best-First Search, A*, and a high-level Double DQN
- Designed a masked macro-action environment with CNN-based state encoding, deadlock pruning, and reward shaping
- Implemented an A*-guided imitation-learning pipeline with expert demonstration generation, and behavior-cloning pretraining, adaptive failed-map sampling

Search Engine | Python, React

Jan 2026 – Mar 2026

- Built a search engine with a custom web crawler, SPIMI-based inverted index, and query-time ranking over crawled web pages
- Implemented polite crawling with URL normalization, duplicate detection, and crawler-trap handling
- Ranked results using TF-IDF, cosine similarity, PageRank, and soft-conjunction retrieval for multi-term queries

Clothing Recommendation | TensorFlow, Flask, React, Google Cloud Storage | GitHub

Aug 2024

- Built an image retrieval system using ResNet-50 embeddings and cosine similarity to return top matches across 10,000+ items
- Developed a Flask API and React interface for querying and displaying recommendations, with image assets served from Google Cloud Storage

HONORS & AWARDS

ICS Honors Program (selective honors program for top-performing CS undergraduates) | S and S International Fund (UCI institutional scholarship recognizing international students for academic excellence and community engagement)

TECHNICAL SKILLS

Proficient: Python, Git, SQL, MySQL, PyTorch, Linux

Familiar: Java, C++, JavaScript, MATLAB, HTML/CSS, AWS (EC2, S3), GCP, React, Flask, Docker, Kubernetes